Capstone Project 2 | Data Visualization & Analytics

Retail / E-Commerce | India | 2024-01-01 to 2024-03-30

TOTAL REVENUE	ORDERS ANALYSED	AVG ORDER VALUE	AVG RATING
Rs.74.84 Cr	10,000	Rs.74,839	3.01 / 5

Institute: Newton School of Technology
GitHub: https://github.com/ARXNDEV/capsstoneee
Tableau Public:
https://public.tableau.com/app/profile/saad.arqam/viz/Book1_17773949686390/Dashboard1?publish=yes
Submission Date: 2026-04-28

Team Member	Role
Arun	Dataset Sourcing + ETL & Cleaning
Harsh Vardhan Singh	EDA + KPI Framework
Saad Arqam	Statistical Analysis

Anmol	Tableau Dashboard
Member 5 (TBD)	Report Writing + PPT & Viva

2. Executive Summary

This report presents an end-to-end analytics study of 10,000 e-commerce transactions across India's five major metropolitan markets — Pune, Chennai, Delhi, Mumbai, and Bangalore — spanning 2024-01-01 to 2024-03-30. The analysis addresses two core business questions: which product categories and cities drive the highest revenue, and what factors most influence customer satisfaction ratings.

The dataset spans four product categories (Clothing, Electronics, Furniture, and Groceries) with transactions distributed remarkably evenly across both categories and cities. Total platform revenue for the period stands at Rs.74.84 Cr with an average order value of Rs.74,839. Customer ratings average 3.01 out of 5, suggesting significant room for satisfaction improvement.

Key Findings at a Glance

Revenue Parity	All four categories contribute within 1.4% of each other (24.3%–25.7%), indicating no single category dominates.
Pune Leads Revenue	Pune generates Rs.16.18 Cr, 8.1% above the city average of Rs.14.97 Cr.
Delivery Speed Irrelevant to Ratings	Pearson correlation between delivery_days and rating = -0.007, statistically insignificant.
Balanced Payment Adoption	Card (31.8%), COD (31.7%), and UPI (31.5%) are used with near-perfect parity.
January Peak	January achieves the highest monthly revenue at Rs.25.65 Cr, 7.0% above the 3-month average.
Missing Data Risk	5% of orders (500) lack payment method data, creating analytics blind spots.

3. Sector & Business Context

3.1 Sector Overview & Current Challenges

The Indian e-commerce market is one of the world's fastest-growing, with platforms competing fiercely on price, delivery speed, and product variety. Understanding revenue concentration and customer satisfaction drivers is essential for strategic resource allocation — whether in marketing, logistics, or category management.

3.2 Decision-Maker & Business Value

This project is designed for an executive decision-maker (CEO/COO/Head of Marketplace) responsible for revenue growth and customer experience. Solving the problem supports better investment decisions across categories and cities, and reduces churn risk driven by weak satisfaction scores.

4. Problem Statement & Objectives

4.1 Formal Problem Definition

Business Question: Which product categories and cities drive the highest revenue, and what operational factors (delivery speed, payment method, price point) most influence customer satisfaction ratings across Indian e-commerce transactions?

4.2 Scope & Success Criteria

In scope: ETL in Python, KPI framework definition, EDA and statistical testing, and an interactive Tableau dashboard for decision support. Out of scope: real-time pipelines, customer-level lifetime value modelling, and experimentation (A/B testing). Success is defined as producing a reproducible cleaned dataset, a KPI-driven dashboard, and actionable recommendations supported by evidence.

5. Data Description

Dataset source: course-provided / synthetic transactional dataset. Direct access: https://github.com/ARXNDEV/capsstoneee/blob/main/data/raw/ecommerce_dataset.csv

Attribute	Detail
Source	Course-provided / synthetic transactional dataset (representative of Indian e-commerce)
Size	10,000 rows × 12 columns (raw)
Period	2024-01-01 to 2024-03-30
Sector	Retail / E-Commerce
Categories	Clothing, Electronics, Furniture, Groceries
Cities	Bangalore, Chennai, Delhi, Mumbai, Pune
Null Values	1,500 missing cells across raw data

Quality Issues	Inconsistent city casing ('pune' vs 'Pune'), missing delivery days, missing payment methods, missing p

6. Data Cleaning & ETL Pipeline

6.1 ETL Pipeline Overview (Python)

The full ETL pipeline is implemented in Python using Jupyter Notebooks and a standalone scripts/etl_pipeline.py module. Every transformation is logged and version-controlled in GitHub. The raw dataset in data/raw/ is never modified. Reference: notebooks/02_cleaning.ipynb.

Step 1: Extract: Loaded raw CSV (10,000 rows, 12 columns) from data/raw/ecommerce_dataset.csv using pandas read_csv.

Step 2: City Normalisation: Applied str.strip().str.title() to standardise 'pune' (200 rows) to 'Pune', consolidating all Pune records correctly.

Step 3: Null — product_name: Filled 500 missing product_name values with 'Unknown' to preserve row integrity without introducing bias.

Step 4: Null — payment_method: Filled 500 missing payment_method values with 'Unknown'; these are flagged separately in analysis.

Step 5: Null — delivery_days: Filled 500 missing delivery_days with the column median (5.0 days) — appropriate for near-normally distributed data.

Step 6: Date Parsing: Converted order_date string to datetime; extracted month number, month_name, and ISO week number.

Step 7: Feature Engineering: Created price_per_unit, is_high_value (top 10%), fast_delivery (<=3 days), high_rating (>=4) flags.

Step 8: Load: Saved cleaned data to data/processed/ecommerce_cleaned.csv (10,000 rows, 16 columns). Also exported KPI aggregation tables for Tableau.

6.2 Missing Values, Outliers, Types, Assumptions

Column	Issue	Null Count	Treatment	Rows Affected
product_name	Missing values	500	Fill: 'Unknown'	5.0%
payment_method	Missing values	500	Fill: 'Unknown'	5.0%
delivery_days	Missing values	500	Fill: median (5.0)	5.0%
city	Inconsistent casing	0	Normalise to title case	200 rows
All other columns	No issues	0	None required	–

7. KPI & Metric Framework

The following KPIs were designed to directly address the business question and drive actionable dashboard design for operations, marketing, and customer experience teams.

KPI	Formula	Value	Business Use
Total Revenue	SUM(total_amount)	Rs.74.84 Cr	Overall platform performance
Avg Order Value (AOV)	MEAN(total_amount)	Rs.74,839	Pricing & upsell strategy
Revenue Share by Category	Cat Revenue / Total	24–26%	Category investment decisions
Avg Customer Rating	MEAN(rating)	3.01 / 5.0	Service quality benchmark
Avg Delivery Days	MEAN(delivery_days)	5.0 days	Logistics performance
Fast Delivery Rate	COUNT(delivery_days<=3) / N	~22%	Premium delivery expansion
High-Value Order Rate	COUNT(top 10%) / N	10%	VIP customer targeting
High-Rating Rate	COUNT(rating>=4) / N	~40%	Satisfaction improvement target

8. Exploratory Data Analysis (EDA)

5.1 Revenue Distribution by Category

Revenue is distributed with remarkable parity across all four categories. Furniture leads at Rs.19.24 Cr (25.7%), followed by Groceries at Rs.18.80 Cr (25.1%), Electronics at Rs.18.64 Cr (24.9%), and Clothing at Rs.18.16 Cr (24.3%). The maximum spread is only 1.4 percentage points, confirming no single category dominates the platform.

Category	Revenue (Rs. Cr)	Orders	Avg Order Value (Rs.)	Avg Rating	Revenue Share
Furniture	19.24	2,559	75,205	3.00	25.7%
Groceries	18.80	2,497	75,289	3.01	25.1%
Electronics	18.64	2,502	74,495	3.06	24.9%
Clothing	18.16	2,442	74,345	2.98	24.3%

5.2 Revenue Distribution by City

Pune leads all cities with Rs.16.18 Cr, representing 8.1% above the city average of Rs.14.97 Cr. The revenue gap between the highest (Pune, Rs.16.18 Cr) and lowest (Mumbai, Rs.14.37 Cr) city is 12.5%, suggesting city-specific demand variations that could be further explored with demographic and seasonal data.

City	Revenue (Rs. Cr)	Orders	Avg Order Value (Rs.)	Revenue Share
Pune	16.18	2,189	73,954	21.6%
Chennai	15.06	1,966	76,625	20.1%
Delhi	14.78	1,946	75,961	19.7%
Bangalore	14.44	1,965	73,476	19.3%
Mumbai	14.37	1,934	74,328	19.2%

9. Statistical Analysis

6.1 Correlation Analysis

Pearson correlations were computed between all numeric variables. The analysis reveals near-zero correlations between delivery_days and rating (r = -0.007), between price and rating (r = 0.004), and between total_amount and rating (r = 0.005). These findings confirm that neither price point nor delivery speed explains customer satisfaction variance in this dataset.

Variable Pair	Pearson r	p-value	Interpretation
delivery_days vs rating	-0.0073	0.471	Negligible — delivery speed not linked to satisfaction
price vs rating	0.0042	0.679	Negligible — price point irrelevant to rating
total_amount vs rating	0.0051	0.614	Negligible — order value irrelevant to rating
quantity vs total_amount	0.6812	<0.001	Strong positive — higher quantity = higher revenue (expected)

6.2 One-Way ANOVA: Rating across Categories

A one-way ANOVA was conducted to test whether average customer ratings differ significantly across the four product categories. Results: $F(3, 9996) = 1.462$, $p = 0.2227$. Since $p > 0.05$, we fail to reject the null hypothesis — there is no statistically significant difference in customer ratings across product categories. All categories perform similarly (2.98–3.06 average).

6.3 T-Test: Pune vs Other Cities Revenue

An independent samples t-test compared average order value in Pune vs all other cities combined. The difference is not statistically significant ($p > 0.05$), suggesting Pune's revenue lead is driven by higher order volume, not higher order values.

6.4 Chi-Square: Payment Method vs High-Value Orders

A chi-square test of independence found no significant association between payment method and likelihood of placing a high-value order (chi2, $p > 0.05$). Card, COD, and UPI users show similar high-value order rates — payment preference is not a predictor of customer spending tier.

Test	Variables	Statistic	p-value	Finding
Pearson Correlation	Delivery vs Rating	r = -0.007	0.471	Not significant
One-Way ANOVA	Rating ~ Category	F = 1.462	0.223	Not significant
One-Way ANOVA	Revenue ~ City	F = 2.1	0.079	Not significant
Chi-Square	Payment vs High-Value	chi2 = 1.8	0.402	Not significant
T-Test	Pune vs Others AOV	t = 0.92	0.358	Not significant

10. Tableau Dashboard Design

10.1 Dashboard Objective

The dashboard is designed to help leaders monitor revenue performance and customer satisfaction at a glance, then drill down by city, category, payment method, and delivery performance to diagnose issues.

10.2 View Structure

Executive View: total revenue, AOV, average rating, and trend lines over time. Operational View: breakdown by city and category, rating distribution, delivery-day distribution, and payment mix.

10.3 Filters & Interactivity

Filters include city, category, month, and payment method. Users can click bars/points to cross-filter related charts.

10.4 Tableau Public URL

Tableau Public URL: https://public.tableau.com/app/profile/saad.arqam/viz/Book1_17773949686390/Dashboard1?publish=yes

Reference: tableau/dashboard_links.md and tableau/screenshots/ in GitHub.

11. Insights Summary

1 **Category Revenue Parity**
All four categories contribute similar revenue share, so growth requires differentiation strategy.

2 **Pune Leads Revenue**
Pune leads total revenue mainly via higher order volume, not higher average order value.

3 **Satisfaction Is Platform-Wide**
Average rating is ~3/5 and not strongly explained by delivery speed, price, or category.

4 **Payment Mix Is Balanced**
Card/COD/UPI show near parity, indicating broad payment adoption and opportunity to reduce COD costs.

5 **January Revenue Peak**
January is higher than the Q1 average, indicating seasonality that impacts inventory and operations.

6 **Data Completeness Gaps**
Missing payment and delivery fields create analytics blind spots; validation is required.

7 **High-Value Segment Exists**
Top 10% high-value orders are a target for VIP retention programs.

8 **Operational Levers Beyond Delivery**
Given weak link between delivery days and ratings, focus should shift to product quality and post-purchase experience.

12. Recommendations

REC 1
High

Launch Category-Specific Premium Collections

Since all categories generate near-equal revenue, introduce premium tiers (Electronics Pro, Designer Clothing, Premium Furniture) to capture higher-AOV orders and shift revenue concentration toward high-margin products.

Expected Impact: Revenue growth, AOV increase

REC 2
High

Double Down on Pune With Loyalty Programme

Pune already leads in revenue (Rs.16.18 Cr). Launch a city-specific loyalty programme offering double reward points, early sale access, and free premium delivery to retain Pune's top customers and grow their share of wallet.

Expected Impact: Revenue retention, LTV increase

REC 3
Critical

Fix Customer Satisfaction Through Product Quality Audits

A platform-wide average rating of 3.01 is a serious risk to retention. Since delivery speed doesn't drive ratings, conduct product quality audits in Clothing and Furniture (lowest ratings: 2.98 and 3.00). Enforce minimum quality standards for sellers and implement post-delivery feedback loops.

Expected Impact: Customer retention, NPS improvement

REC 4
Medium

Incentivise Digital Payments to Reduce COD Costs

COD accounts for 31.7% of orders, incurring additional logistics costs. Offer cashback or reward points for UPI/Card payments. Each 5% shift from COD to digital saves approximately Rs.15–25 per order in handling costs.

Expected Impact: Cost reduction, payment data completeness

REC 5
Medium

Implement Real-Time Data Completeness Checks

5% of orders lack payment and delivery data, creating analytics blind spots. Implement mandatory field validation at checkout (payment_method required) and delivery confirmation triggers to ensure 100% data completeness going forward.

Expected Impact: Analytics reliability, operational visibility

13. Impact Estimation

Impact estimates are directional and based on conservative assumptions. The goal is to support prioritisation decisions.

Recommendation	Assumption	Estimated Impact
Shift 5% from COD to digital	Savings ■15–■25 per order for 5% of orders	■0.08–■0.13 Cr per quarter
Improve satisfaction (rating) by +0.2	Retention lift +1–2% on repeat customers	Higher LTV; requires real customer history to quantify
VIP program for top 10%	Increase repeat rate +5% for high-value segment	Revenue uplift; dependent on campaign uptake
Data completeness to ~100%	Reduce missing operational signals to near-zero	Improves decision accuracy; reduces dashboard risk

14. Limitations

- The dataset covers only Q1 2024 (January–March). Seasonal variations across all four quarters cannot be assessed. Year-on-year comparison is not possible.
- 5% missing values in payment_method and delivery_days may introduce slight bias in those specific analyses.
- The dataset is course-provided/synthetic and representative — real-world distributions may show stronger correlations with external factors (promotions, festivals, economic conditions).
- No customer demographics (age, gender, income tier) are available, limiting segmentation depth.
- Returns and refund data are absent, which would materially affect the true revenue and satisfaction picture.

15. Future Scope

- Extend analysis to full calendar year 2024 for seasonal trend modelling (Diwali, end-of-season sales).
- Integrate customer demographics to enable cohort analysis and personalised recommendation modelling.
- Build a churn prediction model using rating history and purchase frequency.
- Implement real-time Tableau dashboard with live database connection for operational monitoring.
- Conduct A/B test on delivery incentives vs. product quality improvements to measure rating impact.

16. Conclusion

This project demonstrates an end-to-end analytics workflow: defining a business problem, building a reproducible Python ETL pipeline, deriving KPIs, performing EDA and statistical tests, and packaging insights into a Tableau dashboard with clear recommendations. Key value delivered: a decision-ready view of revenue distribution and satisfaction drivers, along with prioritised actions to improve outcomes.

17. Appendix

17.1 Data Dictionary (Key Fields)

Column	Description
order_id	Unique order identifier
customer_id	Customer identifier
order_date	Date of order placement
product_category	Category of product (4 categories)
product_name	Product name (missing filled as 'Unknown')
price	Unit price in INR
quantity	Units ordered
total_amount	price × quantity
payment_method	Card/COD/UPI/Unknown
city	Delivery city (standardised casing)
delivery_days	Days to deliver (missing filled with median)
rating	Customer rating 1–5

Full dictionary reference: docs/data_dictionary.md

18. Contribution Matrix (Mandatory)

This matrix should match evidence in GitHub Insights, PR history, and committed files. Update it after final merges if needed.

Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT & Viva
Arun	Lead	Lead	Support	Support	Support	Support	Support
Harsh Vardhan Singh	Support	Support	Lead	Support	Support	Support	Support
Saad Arqam	Support	Support	Support	Lead	Support	Support	Support
Anmol	Support	Support	Support	Support	Lead	Support	Support
Member 5 (TBD)	Support	Support	Support	Support	Support	Lead	Lead

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This project demonstrates end-to-end analytical capability:
Python ETL Pipeline | Statistical Analysis | Tableau Dashboard | Business Storytelling

Tools don't get you marks. Thinking gets you marks.